Melisa Araç

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Summary

Computer Engineering student with a strong interest in **embedded systems, Edge AI, and intelligent embedded applications**. Experienced in **CAN bus communication, DBC-based signal processing, ECU simulation, digital logic design, and machine learning methods such as anomaly detection and regression**. Passionate about combining **low-level programming, real-time systems, and data-driven approaches** to develop reliable and innovative engineering solutions.

Education

Marmara University 2022 – 2026
Bachelor of Science in Computer Engineering

TED Afyon College 2021
Full Education Scholarship

Experience

Long-Term Embedded Software Intern, Baykar Technologies Feb 2026 – Present

- Contributing to embedded software development and debugging in a professional aerospace/defense environment.
- Working with low-level system behavior, software integration, and debugging/tracing workflows.
- Gaining hands-on experience in embedded diagnostics, communication, and software validation processes.

Embedded Software Intern, Anadolu Isuzu Jun 2025 – Aug 2025

- Gained hands-on experience in automotive communication systems, including **CAN protocols, DBC file handling, and ECU simulation**.
- Conducted **signal analysis, preprocessing, and interpretation** under varying operating conditions.
- Investigated **AI-based approaches**, including anomaly detection and predictive modeling, for vehicle signal data.
- Strengthened Python, embedded systems, and machine learning integration skills while collaborating with engineers on applied projects.

Projects

Edge AI-Based Smart Home Intrusion Detection System Ongoing
Senior Capstone Project | Raspberry Pi 4, Python, Transformer, Autoencoder, IoT

- Developing an **Edge AI-based intrusion detection system** for smart home networks.
- Designing anomaly detection pipelines using **Transformer and Autoencoder models**.
- Processing IoT traffic data on a **Raspberry Pi edge device** for real-time monitoring and analysis.
- Exploring lightweight and deployable machine learning approaches suitable for **edge inference**.

CAN Bus ECU Simulator & Signal Analysis System 2025
Python, python-can, cantools, DBC

- Developed a **Python-based ECU simulator** capable of generating and receiving CAN messages based on DBC definitions.
- Parsed **DBC files** to dynamically extract message and signal structures.
- Implemented automated signal generation, CAN frame transmission, and signal-level processing workflows.
- Performed preprocessing and interpretation of signal data to support further analytics and anomaly detection studies.

Machine Learning Applications 2025
IBM Machine Learning Professional Certificate | Python, NumPy, scikit-learn

- Implemented regression models both from scratch and using Python ML libraries.
- Applied supervised and unsupervised learning techniques on real-world datasets.
- Built anomaly detection workflows and evaluated model performance.

Signal Modulation and Demodulation 2025
Python, DSP, FFT, Digital Filtering

- Developed a system to demodulate composite audio signals sampled at **96 kHz**.
- Applied **FFT and digital filtering** techniques to separate two speech signals modulated around 30 kHz and 35 kHz.
- Performed signal reconstruction and analysis using digital signal processing methods.

Custom 20-bit CPU Architecture 2024
Logisim, C

- Designed and implemented a custom **20-bit CPU** with a 12-bit address space and custom instruction set.
- Built core modules including **ALU, Control Unit, Register File, Instruction Memory, and Data Memory**.
- Defined instruction behavior for arithmetic, logical, memory, and branching operations.
- Developed a **custom assembler in C** to translate assembly instructions into machine code for the processor.

Simple Shell Implementation 2024
C, Linux

- Created a basic shell program supporting foreground and background process execution.
- Implemented command parsing and job control using **fork()**, **exec**, and **wait()** mechanisms.
- Managed process behavior and signal handling in a Linux environment.

Skills

Programming: C, Python

Embedded Systems & Low-Level: Embedded C, Digital Logic Design, Processor Architecture, ECU Simulation, ARM-based systems, Debugging & Tracing

Automotive & Communication: CAN Bus, DBC File Handling, python-can, cantools, Signal Processing, Message/Signal Analysis

Machine Learning & Data: Regression, Classification, Anomaly Detection, Supervised Learning, Unsupervised Learning, NumPy, Pandas, scikit-learn, Matplotlib

Tools: Git, Linux, Logisim

Relevant Coursework

Operating Systems, Machine Learning, Computer Architecture, Digital Logic Design, Signal Processing, System Programming

Certificates

IBM Machine Learning Professional Certificate 2025

Languages

Turkish (Native), English (Professional Working Proficiency)